Harrison F. Squires

Experimental Biophysics PhD Student, University of Edinburgh

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Research Experience

Experimental Biophysics PhD, University of Edinburgh *September 2026 – Present*

- Project: "Towards machine olfaction, a new ISR modality", supervised by Prof. Davide Michieletto (Academic) and Dr Neal Hopkins (Industry; Dtsl).
- Research focus: nucleic acid nanotechnology.
- Recipient of SPADS CDT doctoral funding.
- Awarded an additional £10k annual Research Training Support Grant (RTSG) from the OCSA Bio Tech Centre.

University of Edinburgh *June 2026 – August 2026*

Bridging Research Intern

- Awarded bridging funding to undertake research prior to commencing PhD studies.
- Project: "DNA Nanostars", supervised by Prof. Davide Michieletto and Dr Jenny Harnett.
- Investigated liquid–liquid phase separation in DNA nanostars and their potential application in Lanthanide capture.

MPhys Project, University of Edinburgh *September 2025 – May 2026*

- Project: "Evaporation of Breath Figures", supervised by Dr. David Fairhurst and Dr. Joey Kilbride.
- Extended a recently published study by investigating the dynamics of the "macroscopic edge effect" in respiratory breath figures and its relation to droplet size distribution.
- Designed and built an experimental set-up for imaging large droplet arrays.
- Developed image analysis workflows using FIJI and Python.
- Produced efficient scripts for data and statistical analysis.

QinetiQ / University of Edinburgh *June 2025 – August 2025*

Summer Research Intern

- Project: "Terahertz Methods for the Characterisation of Materials". Conducted a literature-based review aimed at advising the companies future research direction.
- Summarised findings in a final report and developed a comparative table of practical trade-offs.
- Funded by the University of Edinburgh through the Career Development Scholarship; project hosted by QinetiQ.

University of Melbourne / Carbon Cybernetics *July 2024 – June 2025*

Research Intern

- Project: "On the characterisation of carbon fibre electrodes for neural recording", supervised by Prof. Steven Prawer.
- *Modelling of Intracortical Microwire Electrodes for Brain–Machine Interfaces.* Manuscript in preparation. Contributing author.
- Designed and built a bench-top testing system to optimise carbon fibre Micro Electrode Array (MEA) performance.
- Undertook a meta-analysis of *in vivo* datasets to inform optimal MEA design.
- Automated data acquisition and analysis workflows using MATLAB and Python.
- Delivered weekly research presentations to the group, incorporating feedback to guide further development.

Education

The University of Edinburgh *2021 – 2026*
Master of Physics (MPhys), Physics with a Year Abroad

- Awarded First-Class Honours (75%).
- Received the Margaret Campbell-Scott Scholarship for outstanding entry qualifications.
- Awarded the Career Development Scholarship for summer project in conjuction with QinetiQ
- Achieved Certificates of Merit for academic excellence in my first two years.
- Awarded the Junior Honours Class Medal for top performance in my third-year cohort.
- Earned the Edinburgh Award for volunteering contributions to the University Sports Union (Jiu Jitsu Club).
- Final-year focus: soft condensed matter (experiment and theory), biological physics, and statistical physics.

Employment History

Newman College, University of Melbourne *March 2025 – June 2025*
Academic Tutor

- Tutored first-year and second-year Bachelor of Science students at the University of Melbourne in *Foundations of Physics (PHYC10009)* and *Quantum and Thermal Physics (PHYC20012)*.
- Ran individual consultations and group tutorials.
- Provided personalised academic support, tailoring sessions to address students' specific areas of difficulty.
- Designed original resources and problem sets to help students engage with physics concepts in novel contexts.

IJ Mellis *June 2023 – December 2023*
Cheesemonger

- Independently managed shop operations, including opening and closing responsibilities.
- Completed specialised training in product knowledge and food safety.
- Developed an expert knowledge of artisanal and regional cheeses.

Volunteering

Jesuit Worldwide Learning *March 2025 – June 2025*
English as a Second Language Tutor

- Volunteer with Jesuit Worldwide Learning to support refugees in developing English language skills.
- Provide personalised, one-on-one language support to a student in Kenya through meaningful, conversational engagement.